Nexlora Heavy Ranking Layer: Infrastructure Analytical Review

Azure OpenAI foundation model ranker (gpt-4o, Standard tier) vs. Databricks AI Search (indexed retrieval, Standard tier)

Prepared by: Senior Machine Learning Engineer | For: Technical design review, job listings surface | Source: Nexlora Infrastructure Comparison Summary (synthetic data) | October 17, 2026

Token Consumption Analysis

Each ranking request consumes **86,400 tokens**: 200 candidates at 432 tokens each (420 input tokens plus 12 output tokens), so token cost grows in direct proportion to the candidate set size. Dividing the 500,000 TPM quota by 86,400 tokens per request (500,000 ÷ 86,400 = 5.79) gives a ceiling of **5 complete ranking requests per minute**, which matches the TPM-bound maximum reported in the summary. Serving Nexlora's observed peak of 800 ranking requests per minute would require 69,120,000 tokens per minute (800 × 86,400), roughly 138 times the provisioned quota, leaving a gap of 795 unserved requests per minute. The current TPM quota is therefore **not sufficient**: it covers only about 0.6% of peak demand (5 of 800 requests per minute), and closing the gap through quota increase requests alone would take a roughly 138-fold increase rather than an incremental adjustment.

QPS Limits Evaluation

Converted to a per-minute figure, the Databricks AI Search ceiling of 950 QPS equals **57,000 requests per minute** (950 × 60), about 71 times Nexlora's 800-request peak, so it **meets peak traffic** with wide headroom at its reported 98% recall@10 operating point. On the separate latency test, Databricks AI Search **meets** the 50 ms p95 SLA at 38 ms, while the foundation model ranker **fails** it at 4,200 ms, 84 times the limit, which by itself rules the foundation model ranker out as the heavy ranker for the job listings surface regardless of throughput or cost. The foundation model ranker reports no QPS ceiling, and this should be read as **missing data rather than unlimited capacity**, since the only throughput bound available for it is the TPM-derived ceiling of about 5.79 requests per minute (roughly 0.1 QPS). This asymmetry gap prevents a like-for-like throughput comparison and leaves open whether a quota increase would unlock proportional capacity: if each candidate is scored with its own prompt, as the per-candidate token metrics suggest, peak load implies 160,000 model calls per minute, and an unreported request-rate or concurrency limit could become the binding constraint even after the TPM gap is closed.

Infrastructure Cost Comparison

At peak load, the foundation model ranker costs **$59.44 per minute** ((800 ÷ 1,000) × $74.30) and Databricks AI Search costs **$0.896 per minute**, about $0.90 ((800 ÷ 1,000) × $1.12), which makes Databricks AI Search about **66 times more cost-efficient** at this volume ($59.44 ÷ $0.896 = 66.3), although the foundation model figure is notional because the current quota can serve only 5 of those 800 requests per minute. The Databricks AI Search entry does not specify a node SKU, which is a validity concern because DBU rates vary by compute node type, so the DBU rate behind the $1.12 figure cannot be checked against current Databricks pricing documentation or reproduced independently. As a result, the $1.12 figure, and the 66-fold ratio built on it, must be treated as a **directional estimate rather than a confirmed benchmark**; the direction is robust, since the undisclosed configuration would need to cost roughly 66 times more than assumed to erase the gap, but the exact magnitude cannot be cited as confirmed until the SKU is documented.

> **Databricks AI Search is the stronger choice** for Nexlora's heavy ranking layer at the observed peak of 800 requests per minute and well beyond it (up to its 57,000-per-minute ceiling), under a p95 latency requirement of 50 ms, and wherever cost per request matters at scale, because it is the only architecture that passes the throughput, latency and cost tests on the available evidence. The Azure OpenAI foundation model ranker becomes viable only under narrow conditions where seconds of latency are acceptable and volume stays within its TPM ceiling, such as low-traffic or offline surfaces, cold-start contexts with sparse interaction history, or selective re-ranking of a small final candidate set after Databricks AI Search retrieval. Even in that re-ranking role, scoring only the top 10 candidates would cut consumption to 4,320 tokens per request and raise the TPM-bound ceiling to about 115 requests per minute, but serving the full 800-request peak would still need about 3.46 million TPM (nearly 7 times the current quota) and a latency budget far looser than 50 ms.